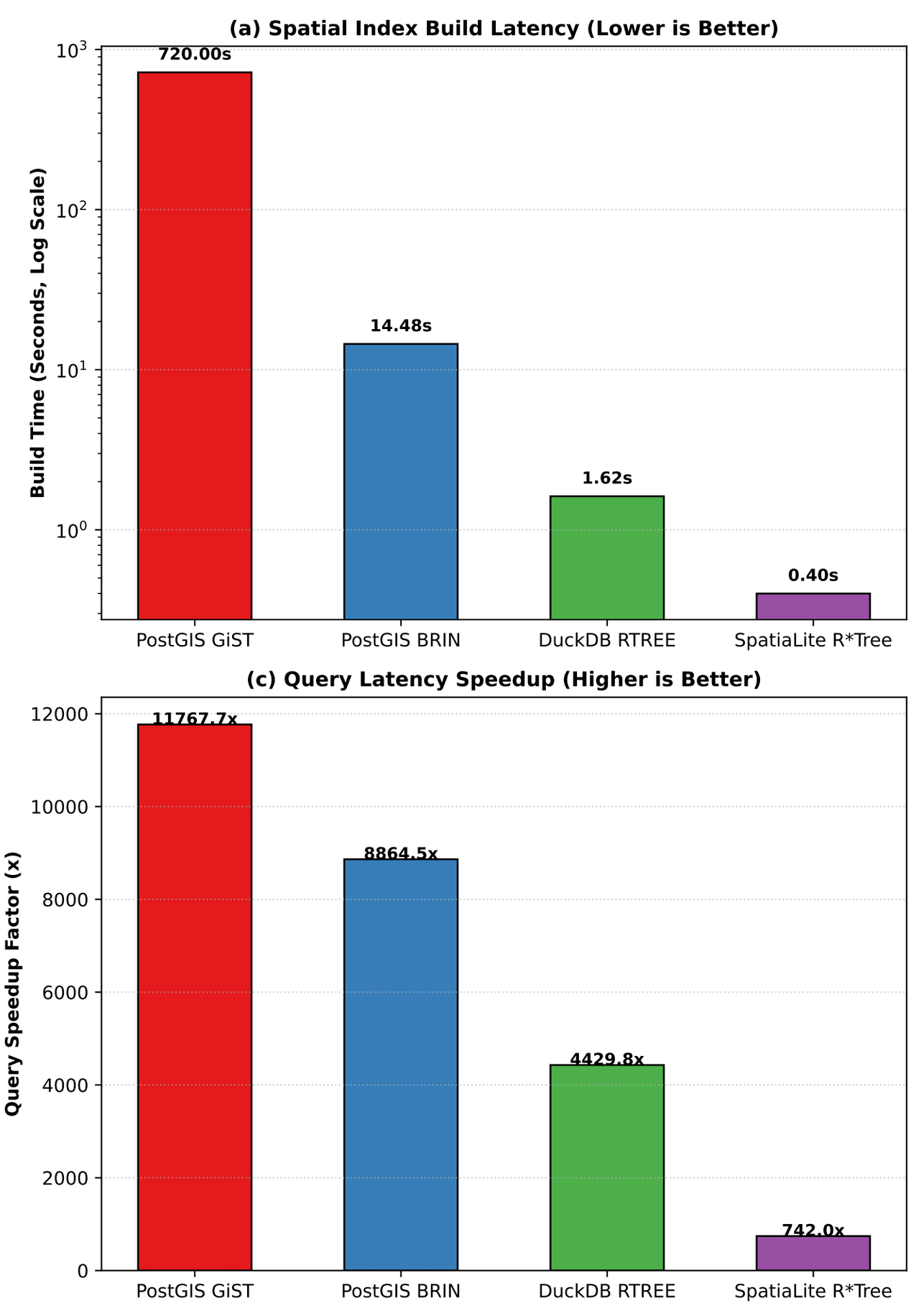
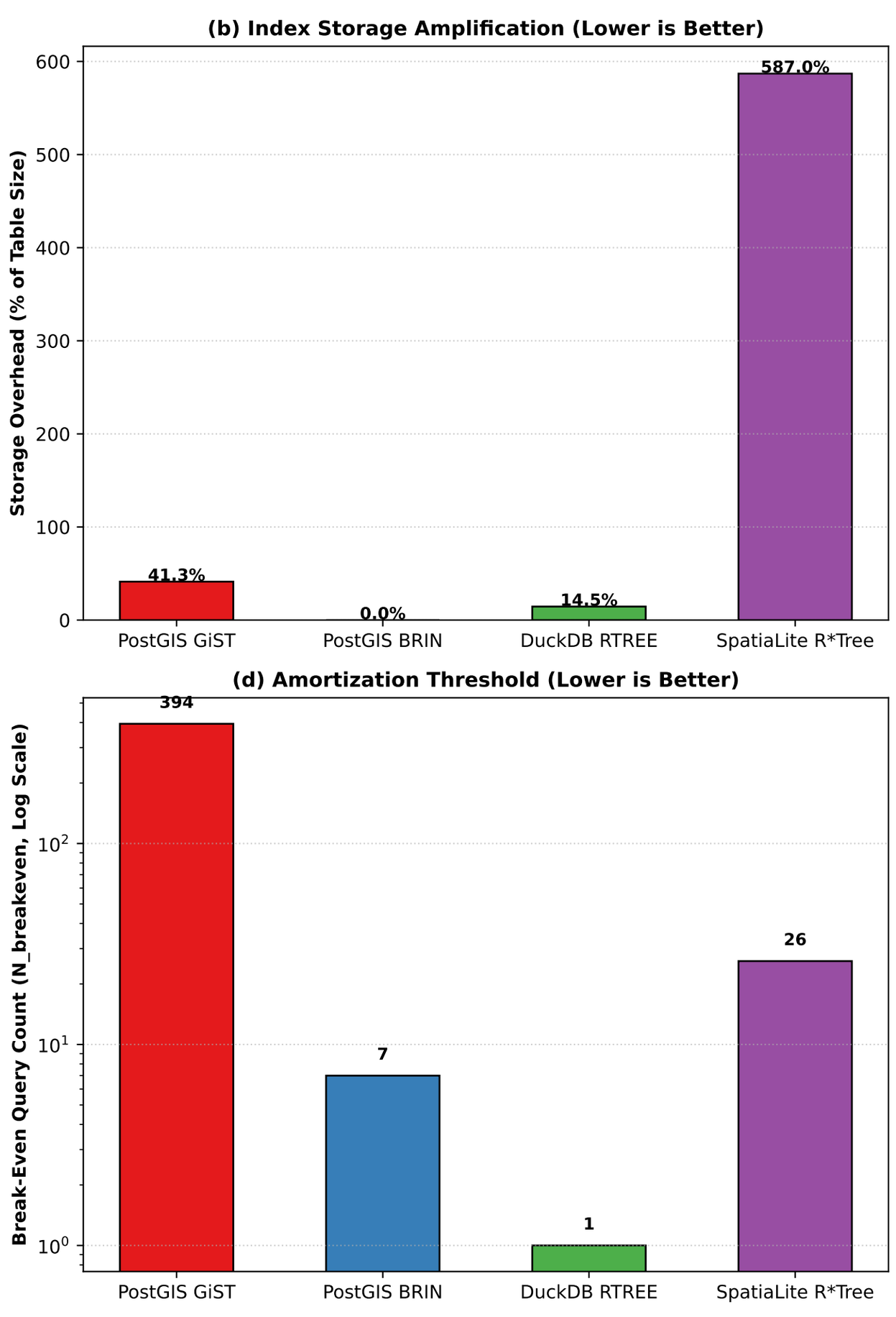

(a) Spatial Index Build Latency (Lower is Better)
Build Time (Seconds, Log Scale)
10^3
10^2
10^1
10^0
720.00s
14.48s
1.62s
0.40s
PostGIS GiST
PostGIS BRIN
DuckDB RTREE
SpatiaLite R*Tree


(b) Index Storage Amplification (Lower is Better)
Storage Overhead (% of Table Size)
600
500
400
300
200
100
0
41.3%
0.0%
14.5%
587.0%
PostGIS GiST
PostGIS BRIN
DuckDB RTREE
SpatiaLite R*Tree

(c) Query Latency Speedup (Higher is Better)
Query Speedup Factor (x)
12000
10000
8000
6000
4000
2000
0
11767.7x
8864.5x
4429.8x
742.0x
PostGIS GiST
PostGIS BRIN
DuckDB RTREE
SpatiaLite R*Tree

(d) Amortization Threshold (Lower is Better)
Break-Even Query Count (N_breakeven, Log Scale)
10^2
10^1
10^0
394
7
1
26
PostGIS GiST
PostGIS BRIN
DuckDB RTREE
SpatiaLite R*Tree